four-way tasks. Although increasing this ablation to 3% consistently leads to further performance decreases, these declines remain substantially milder compared to those observed with induction head ablation for all tasks and settings. This further confirms the importance of specifically induction heads in ICL.

**InternLM2-20B results** The results for InternLM2-20B are presented in Table 1 (bottom). Few-shot prompting consistently outperforms a (close to random) zero-shot baseline across all tasks, providing clear evidence of ICL. The observed patterns are consistent with those seen in Llama-3-8B: Ablating 1% of induction heads leads to a substantial decrease in performance, though these declines are not as pronounced as those seen in the Llama-3-8B model. The trend persists with the ablation of 3% of induction heads, which results in performance decreases of up to 36%. Ablating 3% of randomly selected heads only results in a performance decrease of up to 5.8%. These findings further affirm that induction heads play a crucial role in few-shot ICL, showing that these results generalise across models.

## 6 Ablation Experiments: NLP Tasks

**Datasets and experimental setup** To assess the impact of induction heads on ICL performance in real-world tasks, we conduct ablation experiments on a range of NLP classification tasks. We make use of the following datasets from the SuperGLUE benchmark (Wang et al., 2019): BoolQ (question answering), RTE (natural language inference), WIC (word-sense disambiguation) and WSC (coreference resolution). Additionally, we include ETHOS (Mollas et al., 2020)[3] (hate speech detection), SST-2 (Socher et al., 2013) (sentiment analysis) and SUBJ (Conneau and Kiela, 2018) (subjectivity). All are binary classification tasks.

For all datasets except SST-2, we map the label space to target words "Yes"/"No", as this has been shown to improve ICL performance (Webson and Pavlick, 2022).[4] For BoolQ, ETHOS and SUBJ, we adopt the "Input/Output" template proposed by Wei et al. (2023b). For the other datasets, we use informative prefixes for each element of the prompt (e.g. "Premise:"). Our prompts are shown in App. A.3.2. For all tasks, demonstration examples are selected from the training set and evaluations are performed on the validation set. We report ten-shot prompting performance.

We additionally conduct experiments with semantically-unrelated labels (SUL). Specifically, we map the label space from "Yes"/"No" to "Foo"/"Bar". This removes the semantic priors typically provided by labels, forcing the model to rely solely on input-label mappings for ICL (Wei et al., 2023b). Thus, we expect the model to strongly rely on induction heads to recognise patterns.

We define *ICL benefit* as the metric quantifying the model's advantage from demonstration examples by measuring the performance difference between the ten-shot and zero-shot settings. For each model, we evaluate the full model's ICL benefit for each task and then quantify the percentage change in these benefits due to each ablation.[5]

**Llama-3-8B results** We observe ICL benefits for all tasks except WSC, which we exclude from further analysis. The impact of ablations on these benefits are detailed in Figure 4 (top). Ablating 1% of induction heads reduces the ICL benefit considerably more than a 1% random ablation across all tasks except ETHOS. Specifically, in the SUBJ task, this targeted ablation results in a 63.8% reduction, whereas the random ablation leads to only a 11.8% decrease. For SST-2, we observe a decline of 113% with targeted ablation, indicating that ten-shot performance has dropped below zero-shot level. This suggests that the model may no longer leverage demonstration examples effectively for this task. Increasing the induction head ablation to 3% further decreases the ICL benefit for some tasks.

We further find that for the induction head ablations, the decrease in ICL benefit is due to ten-shot performance being affected more than zero-shot performance for all tasks except ETHOS (App. A.4). Such a pattern is not present in random head ablations. This supports the importance of induction heads for ICL specifically, as opposed to the general functioning of the LLM.

In the SUL setting, the model achieves above-random performance only for the ETHOS, SST-2 and SUBJ tasks, which are presented in Figure 5 (top). Ablating 1% of induction heads leads to substantial performance declines of around 20% for all tasks. For random ablations at the same level, these declines are considerably smaller. These findings

[3] We employ a 80/20% training-validation split.

[4] For SST-2, we found this to actually lead to poor zero-shot performance, so we use the labels "Positive"/"Negative".

[5] A full overview of model accuracy is shown in App. A.4.

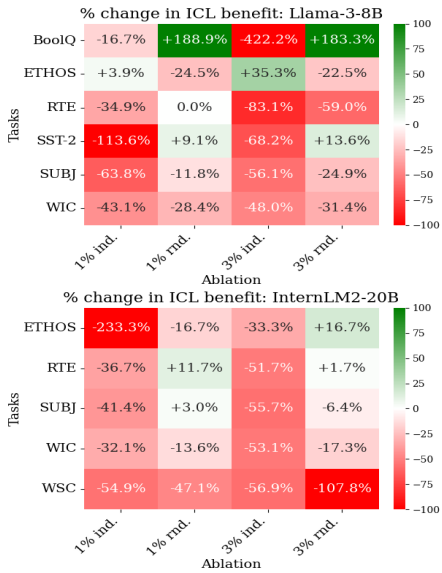


Figure 4: Change in ICL benefit for Llama-3-8B (top) and InternLM2-20B (bottom), due to head ablations when compared to that of the full model. "1% ind." and "3% ind." denote ablating the top respective percentage of induction heads. "1% rnd." and "3% rnd." denote randomly ablating the respective percentage of all heads in the model.

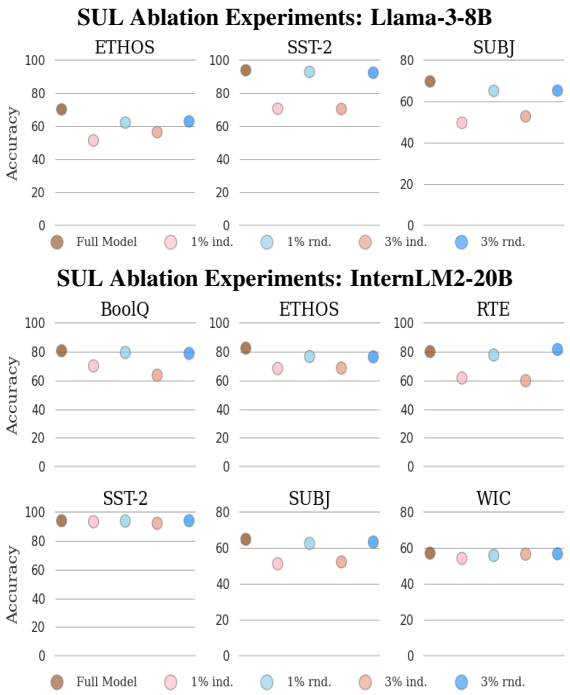


Figure 5: SUL ablation experiments for Llama-3-8B (top) and InternLM2-20B (bottom) on NLP tasks. "1% ind." and "3% ind." denote ablating the top respective percentage of induction heads. "1% rnd." and "3% rnd." denote randomly ablating the respective percentage of all heads in the model.

suggest that the model's capacity to reference previous examples for learning input-label mappings may be compromised. Increasing the induction head ablation to 3% did not reduce performance further, indicating that the heads responsible are amongst those with the highest prefix-scores.

**InternLM2-20B results** We observe ICL benefits on all tasks with the exceptions of BoolQ and SST-2, which we therefore exclude from further analysis. The results for the remaining tasks are shown in Figure 4 (bottom). Consistent with observations from the Llama-3-8B model, we note that: (1) targeted ablations of induction heads negatively affect ten-shot performance more than zero-shot performance across all tasks, whereas random ablations do not follow this pattern (App. A.4); (2) a 1% targeted ablation of induction heads diminishes the ICL benefit considerably more than a 1% random ablation. Unlike in Llama-3-8B, increasing the targeted ablation to 3% further reduces the ICL benefit in all tasks other than ETHOS. For WSC, the 3% random ablation appears to decrease the ICL benefit more than the targeted ablation. However, this effect is partly attributed to an increase in zero-shot performance from the random ablation (App. A.4).

In the SUL setting, the model achieves above-random performance in all tasks except for WSC and the results are presented in Fig. 5 (bottom). Induction head ablations consistently lead to performance declines of up to 20%, which are generally substantially larger than those caused by random ablations across tasks, although the differences are smaller for SST-2 and WIC.

We observed similar trends for the five-shot setting (App. A.4). These consistent findings across different models and settings reaffirm our conclusions on the critical role of induction heads in ICL.

## 7 Attention Knockout Experiments

**Attention pattern analysis** To further explore the functional significance of induction heads, we leverage the structure of the word-sequence datasets. For instance, in the binary WORDSEQ 1 task (classifying <fruit-animal> pairs vs. <furniture-profession> ones), if an induction head focuses on a *furniture* token, it should typically attend to *profession* tokens that have historically followed *furniture*. To validate this, we conduct a qualitative analysis of the attention patterns exhibited by the top five induction heads with the highest prefix matching scores from Llama-3-8B on the binary and four-way versions of the WORDSEQ 1

Figure 6: Attention pattern for the final ":" token in head 30 of layer 15 in the Llama-3-8B model.

Figure 7: Attention pattern for the "ape" token in head 30 of layer 15 in the Llama-3-8B model.

task (App. A.6).

Figure 6 illustrates the attention pattern of the highest scoring induction head for the final ":" token on a sample from the binary task. This token predominantly attends to label tokens, particularly focusing on "Foo," the target label in this instance. Such patterns, while not universally consistent, are particularly evident in induction heads located in later layers, suggesting that induction mechanisms may facilitate the model's ability to selectively focus on relevant labels in certain contexts.

Figure 7 displays the attention pattern for the "ape" token (from 'grape'). We observe that the head attends to tokens that previously followed fruit tokens. This behaviour was consistent across token categories and observed in various heads and samples, suggesting that induction heads not only learn input-label mappings, but may also grasp the underlying pattern itself.

**Experimental setup** To disrupt this induction pattern, we block attention from each token to all tokens that directly followed tokens of the same category, including word categories, labels, newlines, and colons. In contrast, the letter-sequence tasks require recognising abstract patterns rather than categorical sequences. Thus, our blocking strategy is exclusively focused on labels, newlines, and colons.

**Llama-3-8B results** Table 4 (top) shows that blocking the induction pattern for 1% of the induction heads results in performance declines comparable to those observed with complete head ablations. When the block is increased to 3% of the induction heads, performance declines are similarly consistent in most settings. Though in the ten-shot binary word sequence tasks the declines are notably lower, they are still substantial, exceeding 17% for both tasks. These persistent declines across both full and pattern-specific ablations provide strong empirical evidence that induction heads predominantly rely on this attention pattern.

**InternLM2-20B results** Table 4 (bottom) shows that blocking the induction pattern in 1% of induction heads results in performance declines within 4.1% of full head ablations. Unlike in Llama-3-8B, this approach consistently causes greater performance decreases than full ablations. This gap widens further when the knockout is applied to 3% of the induction heads. These observations indicate that for this model, blocking the induction pattern has a more detrimental impact on head functionality than eliminating the head entirely, suggesting that the induction heads are dependent on their specific operational patterns for performance.

## 8 Conclusion

In this paper, we explored the extent to which the prefix matching and copying capabilities of induction heads play a role in few-shot ICL. Our findings highlight a general trend where the ablation of induction heads affects few-shot performance more severely than zero-shot performance. This observation underscores the significance of induction heads in enabling the model to learn effectively from a limited number of examples. Moreover, even a minimal ablation of just 1% of these heads results in a substantial decrease in ICL, suggesting that induction heads are crucial for the model's few-shot learning capabilities. Additionally, when the induction pattern is blocked for a percentage of these heads, performance drops to levels comparable to full head ablations. This provides further empirical evidence that induction heads predominantly utilise this induction pattern and are dependent on it for optimal performance. Overall, our findings indicate that induction heads are a fundamental mechanism underlying ICL.